KARTHIK SUBRAMANIAN

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EDUCATION

University of California, Berkeley *May 2028*
B.S. in Electrical Engineering and Computer Sciences | Minors in Data Science & Public Policy | Regents' and Chancellor's Scholar

PROFESSIONAL EXPERIENCE

Astra Medical *[backed by Berkeley SkyDeck · supported by NVIDIA Inception and Salesforce Launchpad]* **Berkeley, CA**
Co-Founder *Mar 2026 - Present*
- Led the scaling of an AI platform to automate medical billing, targeting 15+ hr/wk reductions in administrative overhead.
- Optimized agentic PDF extraction pipelines to extract medical codes and cut preventable insurance claim denials by 50%.

WorkWhile *[backed by Khosla Ventures, Reach Capital, and Citi Impact Fund]* **San Francisco, CA**
Software Engineering Intern *Apr 2026 - Present*
- Led development of production-grade AI automation infrastructure with the COO for a $50M, WEF-cited labor platform.
- Built agentic workflows via Make, n8n, and custom APIs, streamlining employment pipelines for 1.2M+ frontline workers.

Scam AI *[backed by Berkeley SkyDeck, Llama Ventures, and SCVxSBI]* **Berkeley, CA**
Machine Learning Researcher *May 2026 - Aug 2026*
- Engineered and evaluated deepfake detection models for the world's most accurate multimodal media integrity platform.
- Developed full-pipeline ML systems with the CTO for a $2.6M-backed platform targeting large-scale AI-generated threats.

Git Machine Labs *[backed by Berkeley SkyDeck]* **Berkeley, CA**
GPU Optimization Researcher *May 2026 - Jul 2026*
- Developed vLLM decode roofline attribution pipelines, isolating per-kernel GPU bottlenecks from CUPTI/NVML telemetry.
- Built a validated optimization loop, applying reversible GPU configs without code rewrites toward a 15% throughput floor.

Computer Science Undergraduate Association **Berkeley, CA**
Technology Committee Officer *Oct 2025 - May 2026*
- Maintained Linux servers, LDAP, and email systems; developed internal tools and hosted hackathons for 1,000+ members.
- Held office hours to assist students with Git/GitHub and Python/Java; coordinated talks, workshops, and industry events.

Space Technologies at California **Berkeley, CA**
Software Engineer *Sep 2025 - May 2026*
- Built computer vision systems for a lunar hopper, merging camera feeds, refining imagery, and ensuring spatial alignment.
- Developed autonomous navigation, crater avoidance, and high-fidelity data collection, yielding detailed lunar terrain maps.

CoPatible **Remote**
Lead Software Engineer Intern *Jan 2026 - May 2026*
- Led a cross-functional team of 10 to architect a full-stack MVP and human-AI workflows for an SF-based founder platform.
- Engineered AI-powered matching logic and sentiment analysis for a voice-first social discovery app backed by ElevenLabs.

Associated Students of the University of California **Berkeley, CA**
Technology Operations Associate, Software Developer *Sep 2025 - May 2026*
- Architected data-driven solutions and disaggregation workflows with the CTO for the campus-wide 45,000+ student suite.
- Developed live occupancy metrics and scalable data pipelines for a Library Crowd Meter to optimize study-space efficiency.

Mosi Audio *[SXSW Innovation Award finalist]* **Santa Clara, CA**
Software Developer *May 2024 - May 2026*
- Architected end-to-end Python workflows with the CEO; automated CI/CD pipelines to reduce manual testing time by 30%.
- Engineered firmware enhancements and built an encryption/logging framework for an electronic music creation platform.

Outer Rim Exploration *[backed by AWS Space Accelerator, 500 Global, and Berkeley SkyDeck]* **Berkeley, CA**
Software Engineering and Embedded Systems Intern *Sep 2025 - Feb 2026*
- Assembled and deployed hodoscopes for muography research, integrating IoT devices (SiPMs, gyroscopes) on-site in Idaho.
- Developed real-time C++ firmware and Python pipelines for data post-processing, calibration, and simulation integration.

George Mason University **Remote**
Research Intern *Feb 2024 - Dec 2024*
- Developed AI/ML models to analyze supersonic flow data, investigating coronal mass ejections in a faculty-led research lab.
- Published in the Journal of Student-Scientists' Research; Presented at the 2024 American Geophysical Union Conference.

Student Bono **Fremont, CA**
Founder and Head of Chapter Network, Chief Expansion Officer, Nonprofit Coordination Director *Feb 2022 - Jun 2025*
- Spearheaded initiatives with over 40 non-profit organizations to raise $95,000+ and aid 500,000+ students worldwide.
- Directed the largest global K-12 VR pilot study (160 students, 12 countries) with the Stanford Graduate School of Education.

HONORS & AWARDS

Inventions
- SafeBox (IoT to protect package deliveries, '21): *Finalist at National Invention Convention; Broadcom MASTERS nominee*
- CalmSpace (assistive IoT platform with acoustic triggers for caregivers, '22): *Merit Award at CA Invention Convention*
- AquaMate (embedded IoT hydration-tracking coaster with mobile syncing, '23): *Finalist at National Invention Convention; National Finalist (Top 5) at University of Iowa Jacobson Institute's Innovator Competition; Top Startup Pitch at LaunchX*
- Supernova Classification (algorithmic machine learning classifier, '24): *Merit Award at California Invention Convention*

Quiz Bowl & History Bowl: Co-founded 5 teams; 2x Qualifier, QB Nationals; Qualifier, HB Nationals; 1st Place, California Cup
More Distinctions: AIME (top 2.5% globally in American Math Competition); 4x Qualifier, National Science Bee; 3x Qualifier, National History Bee; 1st in California & 6x State Finalist, National History Day; 2x Team USA Selection, Geography Bee World Championships; Qualifier, International History Olympiad; 2x Finalist, International Astronomy and Astrophysics Competition

Skills: Python, C++, C, Java, Rust, TypeScript, React, PyTorch, CUDA, RAG, OpenCV, SQL, Linux, Docker, Kubernetes, AWS, Git
Interests: Writing, Board Games, Stargazing, Philosophy, Travel, Trivia, History/Geography, Civic Engagement, Film & Television